Lesson 1 — Extra Practice

Understanding Powers and the Exponent Laws

Part A — What a Power Actually Is

1. A **power** is the *whole* expression a^n — not the little number in the corner. For each expression below, state four things: the **power** itself, its **base**, its **exponent**, and the **coefficient** multiplying it. Read the parentheses carefully — they decide what the base is.

(a) -3^2 (b) $(-3)^2$ (c) $-2x^5$

(d) $(-2x)^5$ (e) $\dfrac{5y^3}{8}$ (f) $-\left(\dfrac{3}{4}\right)^6$

2. Decide whether each statement is **true** or **false**. If it is false, write the corrected statement — do not simply write "false."

(a) In $7x^3$ the power is $7x^3$. (b) In -3^2 the base is -3.

(c) In 5^y there is no base, because the exponent is a letter. (d) In $(2cd)^2$ the base is cd.

(e) In the expression m, the exponent is 1 and the coefficient is 1. (f) "The exponent of x^6 is 6" and "the power is 6" say the same thing.

3. Write each of the following as a repeated multiplication, with no exponents left. Then state which of the four — if any — are equal to each other, and say why.

(a) $3ab^4$ (b) $3(ab)^4$ (c) $(3ab)^4$ (d) $(3a)b^4$

Part B — The Fence: $(-a)^n$, $-a^n$, and Coefficients

4. Evaluate without a calculator. Every one of these turns on where the parentheses sit.

(a) -5^2 (b) $(-5)^2$ (c) -5^3 (d) $(-5)^3$ (e) $-(-5)^2$ (f) $-(-2)^4$

5. Simplify each pair, then state in one sentence what the parentheses changed.

(a) $2(cd)^2$ and $(2cd)^2$ (b) $3(xy)^2$ and $(3xy)^2$ (c) $-4(mn)^3$ and $(-4mn)^3$

(d) $-2a^4$ and $(-2a)^4$ (e) $-(2a)^4$ and $(-2a)^4$ (f) $\left(-5^2\right)^3$ and $\left((-5)^2\right)^3$

6. Let $x = -2$. Evaluate each expression. Substitute the value *inside brackets* first, then apply the exponent, then deal with anything outside the fence.

(a) $-x^2$ (b) $(-x)^2$ (c) $-x^3$ (d) $(-x)^3$ (e) $3x^2$ (f) $(3x)^2$

7. Now generalise what questions 4 and 6 showed you. Let a be any number and let n be a positive whole number.

(a) For which exponents n is $(-a)^n = -a^n$? **Explain** using the number of negative factors.

(b) For which exponents n is $(-a)^n = a^n$?

(c) Is there a value of a for which *both* equations hold no matter what n is? Justify.

Part C — The Laws with Symbolic Exponents

8. The exponent laws do not care whether the exponents are numbers or letters. Simplify each of the following, leaving your answer as a single power with a simplified exponent. Assume every base is non-zero.

(a) $x^n \cdot x^{n+3}$ (b) $\dfrac{y^{2m+5}}{y^{m+1}}$ (c) $\left(a^{k+2}\right)^3$

(d) $b^{n-1} \cdot b^{n+1}$ (e) $c^{3n} \cdot c^n \div c^{2n}$ (f) $\dfrac{w^{4n}}{w^{n-2}}$

9. Same idea, with more than one base in play. Assume every base is non-zero.

(a) $\left(x^{2n}y^{n}\right)^{3}$ (b) $\dfrac{\left(t^{n}\right)^{4}}{t^{n}}$ (c) $\left(m^{a}n^{b}\right)^{c}$

(d) $\dfrac{p^{5k}}{p^{5k}}$ (e) $\left(\dfrac{r^{n}}{s^{2n}}\right)^{2}$ (f) $\dfrac{z^{n+4}\cdot z^{n}}{\left(z^{n}\right)^{2}}$

(state the restriction on p)

10. Coefficients and exponents obey different rules in the same expression: the coefficients are *multiplied or divided*, while the exponents are *added or subtracted*. Simplify.

(a) $\left(3x^{2}\right)\left(4x^{5}\right)$ (b) $\left(2a^{3}\right)^{4}$ (c) $\dfrac{20w^{9}}{4w^{3}}$

(d) $\left(\dfrac{2r^{3}}{s^{2}}\right)^{3}$ (e) $\left(-3p^{2}q\right)^{3}$ (f) $\dfrac{\left(5m^{2}\right)^{2}}{5m^{3}}$

Part D — Working Backwards to a Missing Exponent

11. Simplify the left side first, then read off what the missing exponent must be. Show the simplified form before you solve.

(a) $\dfrac{\left(a^{\square}\right)^{3}}{a^{4}}=a^{14}$ (b) $\left(x^{2}\right)^{3}\cdot x^{\square}=x^{12}$ (c) $\dfrac{y^{\square}\cdot y^{5}}{\left(y^{3}\right)^{2}}=y^{12}$

(d) $\left(z^{4}z^{\square}\right)^{2}=z^{22}$ (e) $\left(\dfrac{m^{9}}{m^{\square}}\right)^{4}=m^{20}$ (f) $\dfrac{t^{\square}}{t^{\square}}=1$, with the two boxes *different*. Possible? Explain.

12. Find the whole number n in each case.

(a) $(w^n)(w^n)(w^n)(w^n)(w^n) = w^{45}$

(b) $\dfrac{k^{3n}}{k^n} = k^{16}$

(c) $(y^n)^5 = y^{35}$

(d) $v^{2n+3} = v^{n+9}$

(e) $(c^n)^n = c^{36}$

(f) $\dfrac{d^n \cdot d^n}{d^3} = d^{n+5}$

13. Here the coefficient hides information too. Find *both* unknowns, and say which part of the equation pinned down each one.

(a) $(3x^m)^n = 27x^{12}$

(b) $(2a^p)^q = 32a^{20}$

(c) $\left(kx^3\right)^2 = 49x^6$
(two answers — find both)

Part E — Find the Error

14. Asked to simplify $\left(3x^2\right)\left(4x^5\right)$, a student wrote $\left(3x^2\right)\left(4x^5\right) = 7x^7$.

(a) The exponent 7 is correct. Name, in one sentence, exactly what the student did wrong.

(b) Give the correct answer.

(c) Substitute $x = 1$ into the original and into $7x^7$ to show the two disagree.

15. Asked to simplify $\left(x^3\right)^4$, a student wrote $\left(x^3\right)^4 = x^7$.

(a) Which law did the student reach for, and which one did the question actually call for?

(b) Expand $\left(x^3\right)^4$ as a repeated multiplication and count the factors of x.

(c) Write an expression that *does* simplify to x^7, using x^3 as one of its parts.

16. Three more student answers. For each, decide whether it is right or wrong; if it is wrong, correct it and name the error in a few words.

(a) $(4y)^2 = 4y^2$ (b) $\dfrac{a^8}{a^2} = a^4$ (c) $(-2b)^3 = -8b^3$

Part F — Why the Laws Are True

17. **Explain** why $a^m \cdot a^n = a^{m+n}$ for whole numbers m and n. Argue from repeated multiplication — write out what a^m means, what a^n means, and count the factors in the product. Do *not* simply cite the Product Law; the law is what you are being asked to justify.

18. Two more justifications, each from repeated multiplication.

(a) Explain why $(a^m)^n = a^{mn}$. Say clearly what is being repeated, and how many times, and why that produces a *product* of exponents rather than a sum.

(b) Use the Quotient Law on $\dfrac{a^m}{a^m}$ two different ways to explain why $a^0 = 1$. State the restriction on a and say which step forces it.

19. The Power of a Product Law says $(ab)^m = a^m b^m$. A tempting but false companion would be $(a+b)^m = a^m + b^m$.

(a) Verify $(ab)^m = a^m b^m$ for $a = 3$, $b = 4$, $m = 2$.

(b) Test $(a+b)^m = a^m + b^m$ on the same numbers. What happens?

(c) Explain, using repeated multiplication, why the rule works for a product but not for a sum.

Part G — Stretch

20. No calculator on this page. Rewrite each power so that the two being compared share a **common base**, then compare.

(a) Which is larger, 2^{100} or 4^{50}?

(b) Which is larger, 2^{100} or 8^{33}?

(c) Arrange 2^{100}, 4^{50}, 8^{33} and 16^{25} from least to greatest.

21. Order 2^{30}, 3^{20} and 6^{12} from least to greatest, without a calculator. The exponents have no common base, so match the *exponents* instead.

(a) The three exponents 30, 20 and 12 share a common factor. Use it to write all three numbers as (something)^2.

(b) Evaluate the three bases you get, and state the order.

(c) **Explain** why comparing the bases of equal-exponent powers settles the order.

22. *(Multiple Choice)* Which of the following is equivalent to $\left(2x^3\right)^4$?

A. $2x^{12}$ B. $8x^{12}$
C. $16x^7$ D. $16x^{12}$

23. *(Numerical Response)* If $(x^n)^4 \cdot x^n = x^{45}$, where n is a whole number, then n is _____.
(Record your answer as a three-digit number.)

Lesson 2 — Extra Practice

Simplifying Expressions with Several Exponent Laws

Part A — Three and Four Laws in One Chain

1. Each expression needs the product law, the quotient law *and* the power law before it is finished. Work the coefficients first, then one variable base at a time. Give every answer with positive exponents.

(a) $\dfrac{(4x^3y^2)^2(2xy^4)}{8x^2y^3}$

(b) $(2a^2b)^3(3ab^4)^2$

(c) $\dfrac{(-3m^2n^3)^3}{9m^4n^5}$

(d) $\dfrac{(5p^4q)^2(2pq^3)^3}{100p^6q^5}$

(e) $\left(\dfrac{6u^5v^2}{3u^2v}\right)^3(uv)^2$

(f) $\dfrac{(2c^3d^2)^4}{(4c^2d)^2\cdot c^3d}$

2. Here the outer exponent sits on a **whole quotient**. One answer ends with a fraction coefficient, and others pass through fractions on the way — keep them exact.

(a) $\left(\dfrac{12x^5y^3}{4x^2y}\right)^3$

(b) $\left(\dfrac{-2a^4b^5}{a^2b^3}\right)^4$

(c) $\left(\dfrac{5m^3n^4}{10mn^2}\right)^2\cdot 4m^2$

(d) $\left(\dfrac{3r^4s^2\cdot 2rs^5}{9r^3s^4}\right)^2$

(e) $\left(\dfrac{(2x^2y)^3}{4xy^2}\right)^2$

(f) $\left(\dfrac{-4k^5}{2k^2}\right)^3\left(\dfrac{k^2}{2}\right)^2$

3. Simplify to a single power first, then evaluate. Do not reach for a calculator until the exponents are gone.

(a) $\dfrac{(2^3)^4\times 2^5}{(2^6)^2}$

(b) $\left(\dfrac{3^7}{3^4}\right)^2\div 3^4$

(c) $\dfrac{(-5)^6\times 5^3}{(-5)^7}$

(d) $\dfrac{(2^2 \times 3)^3}{2^4 \times 3^2}$

(e) $\left(\dfrac{6^5}{6^3}\right)^2 \div 6^2$

(f) $-\left(\dfrac{4^5 \times 4^2}{4^6}\right)^3$

Part B — Which Order? Brackets First, or Distribute First?

4. Consider $\left(\dfrac{6x^4y^2}{3x^2y}\right)^3$. Both routes below are legal, and both must give the same answer.

(a) **Simplify inside the brackets first**, then apply the outer exponent. Show each line.

(b) **Distribute the outer exponent first** — cube the numerator and cube the denominator — then simplify the quotient. Show each line.

(c) Your two answers should agree. State which route was less work, and say exactly what made it less work.

5. For each expression, decide *first* which route is shorter, write one sentence saying why, then simplify. (One of these has no single bracket to work inside at all — find it.)

(a) $\left(\dfrac{20a^7b^3}{5a^4b}\right)^2$

(b) $\dfrac{(3x^2y^3)^4}{(9x^3y^5)^2}$

(c) $\left(\dfrac{2m^3 \cdot 6m^4}{4m^5}\right)^3$

6. Simplify $\left(\dfrac{4c^5d^2}{6c^3d}\right)^2$ both ways.

(a) Inside the brackets first.

(b) Distributing the outer exponent first.

(c) This time the inside route produces a fraction straight away. Does that make it the worse route? Explain.

Part C — Where the Negative Sign Sits

7. These six are the same quotient six ways: the negative sign moves inside or outside the bracket, and the outer exponent is even or odd. Simplify inside the bracket first, then compare your six answers.

(a) $\left(\dfrac{-4x^5}{2x^2}\right)^2$ (b) $-\left(\dfrac{4x^5}{2x^2}\right)^2$ (c) $\left(\dfrac{-4x^5}{2x^2}\right)^3$

(d) $-\left(\dfrac{4x^5}{2x^2}\right)^3$ (e) $-\left(\dfrac{-4x^5}{2x^2}\right)^2$ (f) $-\left(\dfrac{-4x^5}{2x^2}\right)^3$

8. Now the same idea inside a chain. Watch the sign at every step.

(a) $\dfrac{-(3a^2b)^2}{(-3ab)^2}$ (b) $\dfrac{(-4m^3)^3}{-(2m^2)^3}$ (c) $-\left(\dfrac{-2p^4}{p^2}\right)^3$ (d) $\left(-\dfrac{3x^2}{y^3}\right)^4$

9. In a long chain you can settle the **sign** before you touch a single exponent. Consider

$$\frac{(-2a^3)^5(-3b^4)^2}{-(4a^2b)^3}.$$

(a) Predict whether the simplified answer is positive or negative *without* simplifying. Say how each of the three brackets contributes.

(b) Now simplify the expression completely and check your prediction.

(c) Write the general rule you used as one sentence about counting negative factors.

Part D — Symbolic Exponents in a Chain

10. The laws do not care whether an exponent is a number or an expression — you still add, subtract and multiply the exponents. Simplify each to a single power with the simplest possible exponent.

(a) $\dfrac{(x^{n+2})^3}{x^{2n}}$

(b) $\dfrac{(a^{2k+1})^2}{a^{3k-4}}$

(c) $(y^{m+3})^2 \cdot y^{4-m}$

(d) $\dfrac{(b^{3t})^2 \cdot b^5}{(b^{t+1})^4}$

(e) $\left(\dfrac{c^{2w+3}}{c^{w-1}}\right)^3$

(f) $\dfrac{(p^{n+1})^4}{(p^2)^{2n} \cdot p^{4-2n}}$

11. Each pair below *looks* different. Simplify both sides to a single power and decide whether they are equal for **every** whole number n. If they are not, give one value of n that breaks the pair. (Assume $x \neq 0$.)

(a) $\dfrac{(x^n)^3}{x^n}$ and $(x^n)^2$

(b) $(x^{n+1})^2$ and $x^{2n} \cdot x^2$

(c) $\dfrac{x^{3n}}{x^n}$ and x^3

(d) $(x^2)^n$ and $(x^n)^2$

(e) $x^n \cdot x^n$ and x^{n^2}

(f) $\left(\dfrac{x^{2n}}{x^n}\right)^2$ and $(x^n)^2$

Part E — Find the Error in a Chain

12. Asked to expand $(a+b)^2$, a student wrote

$$(a+b)^2 = a^2 + b^2.$$

(a) Substitute $a = 3$ and $b = 4$ into both sides to show the statement is false.

(b) Expand $(a+b)^2$ correctly and state the term the student lost.

(c) Which law did the student think they were using, and what does that law actually require of what is inside the bracket?

13. Asked to simplify $\dfrac{a^5 + a^3}{a^3}$, a second student crossed out the a^3 in the denominator against the a^3 in the numerator and wrote a^5.

(a) Substitute $a = 2$ into the original expression and into the student's answer. What goes wrong?

(b) Simplify the expression correctly by splitting the quotient into two fractions.

(c) Complete the rule in one sentence: "In a quotient you may cancel a common _____ of the *whole* numerator, never a single _____ of a sum."

14. Three slips that only show up once several laws are in play. For each, name the mistake and give the correct simplification.

(a) $\left(\dfrac{6x^5}{3x^2}\right)^2 = \dfrac{6x^{10}}{3x^4} = 2x^6$

(b) $\dfrac{12m^7n^3}{4m^7n} = 3m^0n^2 = 0$

(c) $\left(\dfrac{a^4}{b^3}\right)^2 = \dfrac{a^4}{b^6}$

Part F — Which Laws Distribute, and Which Do Not?

15. Six statements. Mark each **always true** or **not always true**; for every one you mark "not always true", give a pair of values that breaks it. (Assume no denominator is zero.)

(a) $(ab)^3 = a^3b^3$

(b) $(a + b)^3 = a^3 + b^3$

(c) $\left(\dfrac{a}{b}\right)^3 = \dfrac{a^3}{b^3}$

(d) $\left(\dfrac{a}{b}\right)^3 = \dfrac{a^3}{b}$

(e) $(a - b)^2 = a^2 - b^2$

(f) $(3ab)^2 = 9a^2b^2$

(g) Every statement you marked "always true" has one feature in common, and every false one breaks it. Name that feature, then use it to answer the headline question: is $(ab)^n = a^nb^n$ *always* true?

16. Question 15 settled that $(a+b)^n = a^n + b^n$ is not a law. But is it *ever* true? The answer is yes — in very particular cases, which this question pins down.

(a) Expand $(a+b)^2$ and subtract a^2+b^2. The difference you are left with must equal zero for the statement to hold. What does that force about a and b?

(b) Give one pair of values, not both zero, for which $(a+b)^2 = a^2+b^2$ is true, and explain why a and b can never *both* be non-zero when $n = 2$.

(c) Now try $n = 3$. Expand $(a+b)^3 - (a^3+b^3)$ and factor the result. This time there is a way for both a and b to be non-zero — find a pair that works and check it.

(d) What happens when $n = 1$? Say in one sentence why the $n = 1$ case is not really an exception to anything.

Part G — Stretch

17. An expression that collapses all the way to 1.

(a) Show that $\dfrac{(2x^2y^3)^3(3xy)^2}{72x^8y^{11}} = 1$. Show the numerator before you cancel.

(b) State the restrictions on x and y, and say which exponent law makes the restriction necessary.

(c) Build your own expression equal to 1 whose numerator is $(5m^4n)^2$.

18. Simplify $\dfrac{(4a^3b^2)^2}{8a^4b^4} \cdot \dfrac{b^2}{2a^2}$.

(a) Simplify completely.

(b) One of the two variables has vanished. Which one, and what does that say about the value of the expression?

(c) Show how you could have predicted this by tracking the exponent of that variable alone, without simplifying anything else.

19. *(Multiple Choice)* The simplified form of $\dfrac{(-2x^3y^2)^4}{(4x^2y)^2 \cdot x^2}$ is

A. x^6y^6 B. $-x^6y^6$

C. x^8y^6 D. $\dfrac{1}{4}x^6y^6$

20. *(Numerical Response)* The expression $\left(\dfrac{3a^4b^2 \cdot 2a^2b^3}{a^3b}\right)^2$ simplifies to the form ka^mb^n, where k, m and n are whole numbers. The value of $k + m + n$ is _____.

(Record your answer as a three-digit number.)

Lesson 3 — Extra Practice

Negative Exponents and Scientific Notation

Write every final answer with positive exponents unless the question asks you to evaluate. Assume every variable is nonzero.

Part A — Sign or Placement?

1. A negative exponent and a negative sign are two different things. Before evaluating, state what the exponent is attached to — the whole base, or only the number after the minus sign — then evaluate exactly. Compare (c) and (d) when you are done.

(a) -3^{-2} (b) $(-3)^{-2}$ (c) $(-3)^{-3}$ (d) -3^{-3} (e) $-(-3)^{-2}$

2. The same distinction with a variable factor. Write each with positive exponents.

(a) $-(5x)^{-2}$ (b) $(-5x)^{-2}$ (c) $(-5x)^{-3}$ (d) $-5x^{-2}$ (e) $\left(-5x^{-1}\right)^{-2}$

3. Students often read x^{-3} as "the opposite of x^3." Test that reading. Evaluate each expression at $x = 2$ *and* at $x = -2$.

(a) x^{-3} (b) $-x^3$ (c) $(-x)^3$ (d) $(-x)^{-3}$

(e) Using your table, explain in one sentence what a negative exponent actually does to a base, and why it never changes the base's sign.

4. State whether each statement is true or false. If it is false, write the correct right-hand side.

(a) $-x^{-4} = \dfrac{1}{x^4}$ (b) $(-x)^{-4} = \dfrac{1}{x^4}$ (c) $(-x)^{-5} = -\dfrac{1}{x^5}$ (d) $(3x)^{-2} = 3x^{-2}$

Part B — Crossing the Fraction Bar

5. A factor with a negative exponent moves to the *other* side of the fraction bar and its exponent turns positive — downstairs factors move up just as readily as upstairs factors move down. Every expression below has negative exponents in *both* positions. Write each with positive exponents.

(a) $\dfrac{a^{-3}b^2}{c^4d^{-5}}$ (b) $\dfrac{4m^{-2}n^5}{7p^3q^{-1}}$ (c) $\dfrac{x^{-5}y^{-2}}{x^{-2}y^3}$ (d) $\dfrac{(2c)^{-3}}{(3d)^{-2}}$ (e) $\dfrac{6^{-1}s^{-4}}{3^{-2}t^{-4}}$

6. Now the numerical coefficients carry negative exponents too. Simplify each, writing the answer with positive exponents and the coefficient fully evaluated.

(a) $\dfrac{3^{-2}x^{-4}y^{3}}{3^{-4}x^{2}y^{-1}}$ (b) $\dfrac{-12a^{-5}b^{3}}{18a^{-2}b^{-4}}$ (c) $\dfrac{\left(3m^{-2}\right)^{-3}}{9m^{-5}}$ (d) $\dfrac{2^{-3}k^{4}}{2^{-5}k^{-2}}$

7. Longer chains. Collect the exponents first and move factors across the bar only at the very end.

(a) $\dfrac{14x^{-3}y^{5}}{7x^{2}y^{-1}} \times \dfrac{x^{-4}y^{-2}}{4x^{-6}y^{3}}$ (b) $\left(\dfrac{6a^{-2}b^{4}}{9a^{3}b^{-1}}\right)^{-2}$ (c) $\dfrac{\left(2p^{-3}q^{2}\right)^{-2}}{\left(4p^{2}q^{-1}\right)^{-1}}$

Part C — Fractional Bases

8. Use $\left(\dfrac{a}{b}\right)^{-n} = \left(\dfrac{b}{a}\right)^{n}$ to evaluate each exactly without a calculator. Leave your answers as exact fractions or integers — no decimals.

(a) $\left(\dfrac{2}{3}\right)^{-3}$ (b) $(0.2)^{-2}$ (c) $\left(-\dfrac{1}{5}\right)^{-3}$ (d) $\left(\dfrac{3}{10}\right)^{-2}$ (e) $\left(-\dfrac{2}{3}\right)^{-4}$

9. Flipping the fraction takes the coefficients with it. Simplify, writing the final answers with positive exponents.

(a) $\left(\dfrac{2x}{5}\right)^{-3}$ (b) $\left(\dfrac{3a^{2}}{4b}\right)^{-2}$ (c) $\left(\dfrac{-2m^{-1}}{5n^{3}}\right)^{-3}$ (d) $\left(\dfrac{x^{-2}y}{xy^{-3}}\right)^{-2}$

10. The rule does not care what the base is made of — a whole expression flips exactly like a single letter, provided it is not zero.

(a) $\left(\dfrac{a+b}{c}\right)^{-2}$ (b) $\left(\dfrac{2}{x+1}\right)^{-3}$

(c) Show that $(x+y)^{-1}$ and $x^{-1}+y^{-1}$ are *not* the same, by evaluating both at $x=2$ and $y=3$.

(d) $\left(\dfrac{3}{m^{-1}+n^{-1}}\right)^{-1}$, written as a single fraction.

Part D — Where $a^0 = 1$ Comes From

11. The zero exponent is not a definition pulled out of the air — the quotient law forces it.

(a) Evaluate $7^4 \div 7^4$ directly, without any exponent law. Then apply the quotient law $a^m \div a^n = a^{m-n}$ to the same quotient. What must 7^0 equal?

(b) Repeat the argument with $a^n \div a^n$ for a general base a and a general exponent n, and state the conclusion.

(c) Your argument in (b) divided a^n by itself. **Explain** why that step forces the restriction $a \neq 0$, and state the law with its restriction attached.

12. Evaluate exactly. Watch what each zero exponent is attached to.

(a) $\left(\dfrac{5x^3y^{-2}}{7}\right)^0$ (b) $-3(2a)^0 + (3a)^0$ (c) $\dfrac{\left(4m^2\right)^0 + 4^{-1}}{2^{-2}}$ (d) $\left[-(5.4)^0\right]^{-3}$ (e) $\dfrac{3^0+3^{-1}}{3^{-2}}$

Part E — Find the Error

13. Asked to write x^{-3} with a positive exponent, a student wrote $x^{-3} = -x^3$, reasoning that "the negative comes out front."

(a) Substitute $x=2$ into both sides to show the claim is false.

(b) Give the correct form of x^{-3} with a positive exponent.

(c) In one sentence, say what the student confused a negative *exponent* with.

14. Asked to write $\dfrac{6a^{-4}}{b^{-2}}$ with positive exponents, a student wrote $\dfrac{b^2}{6a^4}$.

(a) The b was handled correctly. Identify, in one sentence, what went wrong with the numerator.

(b) Give the correct answer.

(c) The student's answer is not nonsense — write the expression that *would* simplify to $\dfrac{b^2}{6a^4}$.

15. Each line below contains exactly one error. Identify it and write the correct answer.

(a) $(2x)^{-3} = \dfrac{2}{x^3}$ (b) $\left(\dfrac{a}{4}\right)^{-2} = \dfrac{4}{a^2}$ (c) $\dfrac{1}{3y^{-5}} = 3y^5$

Part F — Stretch

16. Write both sides as powers of the same base, then equate the exponents. Solve for x.

(a) $2^{-x} = \dfrac{1}{32}$ (b) $3^{x} = \dfrac{1}{81}$ (c) $\left(\dfrac{2}{5}\right)^{x} = \dfrac{25}{4}$ (d) $10^{-x} = 0.001$

17. The laws work on symbolic exponents exactly as they do on numbers. Simplify, writing each answer with positive exponents. Take n and m to be integers with $n > 0$.

(a) $\dfrac{x^{-n}}{x^{-3n}}$ (b) $\left(a^{-2n}\right)^{-3}$ (c) $\dfrac{y^{n-2}}{y^{n+3}}$ (d) $4^{m} \times 4^{-m-2}$

18. Simplify $\dfrac{x^{-1} + y^{-1}}{(xy)^{-1}}$ completely. Show every step: rewrite each negative exponent as a fraction first, then combine. State the restrictions on x and y.

19. *(Multiple Choice)* The value of $\dfrac{2^{-2} + 2^{-1}}{2^{-3}}$ is

A. 3 B. 6 C. $\frac{1}{6}$ D. 12

20. *(Numerical Response)* When $\left(\dfrac{3}{4}\right)^{-3}$ is written as $\dfrac{p}{q}$ in lowest terms, the value of $p - q$ is _____.
(Record your answer as a three-digit number.)

Part G — Scientific Notation

21. Write each number in scientific notation, $a \times 10^n$ with $1 \leq a < 10$. Keep every significant figure — nothing here is to be rounded.

(a) 0.00000304 (b) 407 000 000 000 000 (c) 0.05

22. Write each number in standard notation. Watch the negative exponent — it moves the decimal point the other way.

(a) 7.06×10^{-7} (b) 9.003×10^{5}

23. Multiply or divide the coefficients, add or subtract the exponents, and then check that the coefficient you are left with satisfies $1 \leq a < 10$. Write each result in scientific notation.

(a) $(6 \times 10^{-4})(8 \times 10^{-9})$ (b) $(3.6 \times 10^{-5}) \div (9 \times 10^{2})$ (c) $\dfrac{8.1 \times 10^{11}}{2.7 \times 10^{-4}}$

Lesson 4 — Extra Practice

Rational Exponents and Radical Form

No calculator anywhere in this set unless a question says otherwise. Every value below is chosen so that an exact answer is reachable by hand. Read the denominator of each exponent first — it is the index of the root.

Part A — The Denominator Is the Index

1. In $a^{m/n}$ the **denominator n is the index of the root** and the numerator m is the power. For each of the following, first write down which root the denominator calls for, then evaluate exactly.

(a) $64^{2/3}$ (b) $32^{3/5}$ (c) $16^{5/4}$ (d) $243^{2/5}$ (e) $1\,000\,000^{1/6}$

2. A very common slip is to read *every* fractional exponent as a square root — that is, to read $a^{m/n}$ as $\sqrt{a^m}$ no matter what the denominator says. Evaluate each power correctly, then evaluate the mis-read version $\sqrt{a^m}$, and say in one sentence what the mis-read version is actually the value of.

(a) $27^{2/3}$ against $\sqrt{27^2}$

(b) $16^{3/4}$ against $\sqrt{16^3}$

(c) $625^{3/4}$ against $\sqrt{625^3}$

3. Evaluate $64^{5/6}$ **twice**, showing every step both times.

(a) Root first: $\left(\sqrt[6]{64}\right)^5$

(b) Power first: $\sqrt[6]{64^5}$

(c) The two answers agree. State which route you would use by hand, and why.

4. Evaluate each power by taking the **root first**. Then, beside your answer, write down the number you would have had to take a root of if you had done the *power* first.

(a) $4^{7/2}$ (b) $8^{5/3}$ (c) $9^{7/2}$ (d) $25^{5/2}$

Part B — Negative Rational Exponents

5. A negative exponent still means "take the reciprocal," and the denominator still gives the index. Determine the exact value of each.

(a) $16^{-3/4}$ (b) $125^{-2/3}$ (c) $32^{-4/5}$ (d) $10\,000^{-3/4}$ (e) $64^{-5/6}$

6. With a fractional base, flip the fraction first — then the exponent is positive and the index is easy to see. Express each answer as a fraction in lowest terms (or a whole number, where it simplifies to one).

(a) $\left(\frac{8}{27}\right)^{-2/3}$ (b) $\left(\frac{16}{81}\right)^{-3/4}$ (c) $\left(\frac{125}{8}\right)^{-4/3}$ (d) $\left(\frac{1}{32}\right)^{-3/5}$ (e) $\left(\frac{49}{144}\right)^{-1/2}$

7. Now the minus signs are in different places. Decide what each one is attached to before you evaluate.

(a) $\left(\frac{1}{8}\right)^{-5/3}$ (b) $-\left(64^{-1/2}\right)$ (c) $(-8)^{-2/3}$ (d) $\left(-\frac{27}{64}\right)^{-1/3}$ (e) $-(-32)^{-3/5}$

Part C — Converting Both Ways

8. Write each power as an **entire radical** — one radical sign with everything underneath it, in the form $\sqrt[n]{a^m}$.

(a) $x^{4/7}$ (b) $y^{9/2}$ (c) $m^{-7/4}$

(d) $2a^{5/3}$ (e) $(2a)^{5/3}$ (f) $(5c)^{3/2}$

9. Write an equivalent expression **using radicals** — the root taken first, in the form $\left(\sqrt[n]{a}\right)^m$.

(a) $x^{5/8}$ (b) $h^{7/3}$ (c) $k^{-4/9}$ (d) $7^{3/5}$ (e) $(2b)^{-5/6}$

10. Write each power *both* ways — as an entire radical, and using radicals — then evaluate it, and state which of the two forms you could actually finish by hand.

(a) $32^{3/5}$ (b) $144^{3/2}$

11. Run the conversion the other way. Write each radical as a power in the form x^n, where $n \in Q$.

(a) $\sqrt[8]{x^5}$ (b) $\sqrt[3]{x^{11}}$ (c) $\frac{1}{\sqrt[4]{x^9}}$ (d) $\left(\sqrt[5]{x}\right)^{-7}$ (e) $\frac{x}{\sqrt[3]{x}}$

Part D — When the Base Is Negative

12. An **even index over a negative radicand has no meaning** as a real number; an odd index is untroubled by a negative base. State which of the following have no meaning, and evaluate the rest exactly.

(a) $(-49)^{1/2}$ (b) $-49^{1/2}$ (c) $(-27)^{4/3}$ (d) $(-27)^{3/4}$ (e) $(-243)^{3/5}$

13. Now state the rule behind question 12.

(a) **Explain** why $(-125)^{1/3}$ has meaning but $(-125)^{1/2}$ does not.

(b) $(-8)^{2/3}$ and $(-8)^{4/3}$ are positive, while $(-8)^{1/3}$ and $(-8)^{5/3}$ are negative. Say what decides the sign.

(c) Complete: for $a > 0$ and $\frac{m}{n}$ in lowest terms, $(-a)^{m/n}$ has meaning exactly when _____.

14. Assume x represents a positive integer. State which of the following have no meaning, and for the others say whether the value is positive or negative.

(a) $(-x)^{3/8}$ (b) $(-x)^{8/3}$ (c) $-(-x)^{1/4}$ (d) $-x^{1/4}$ (e) $\left(-x^2\right)^{1/2}$

Part E — The Exponent Laws with Rational Exponents

15. The laws do not change when the exponents become fractions — you still add, subtract and multiply them. Simplify each expression. Write your answer as a power with a positive exponent, and then as an entire radical.

(a) $a^{2/3} \cdot a^{5/2}$ (b) $\left(x^{4/5}\right)^{10/3}$ (c) $m^{3/4} \div m^{5/6}$ (d) $\left(\dfrac{c^{3/2}}{c^{1/4}}\right)^2$

16. Coefficients obey their own arithmetic — multiply, divide or raise them as whole numbers while the exponents are added, subtracted or multiplied. Write each answer with positive exponents.

(a) $5p^{3/4} \times 3p^{1/2}$ (b) $\dfrac{12q^{7/3}}{8q^{1/2}}$ (c) $\left(16r^8\right)^{3/4}$ (d) $\left(8s^{-6}t^3\right)^{-2/3}$

17. Each of these collapses to a whole number. Simplify to a single power first, and only then evaluate.

(a) $9^{3/4} \times 9^{1/4}$ (b) $\dfrac{2^{7/3}}{2^{1/3}}$ (c) $\left(5^{2/3}\right)^3$ (d) $8^{1/2} \times 2^{1/2}$ (e) $\dfrac{27^{5/6}}{27^{1/2}}$

18. A root of a root is a single root: convert everything to exponent form, simplify, and write each as a single power.

(a) $\sqrt[3]{\sqrt{x^{12}}}$ (b) $\sqrt{\sqrt[4]{y^{20}}}$ (c) $\sqrt[3]{\sqrt[4]{4\,096}}$ (d) $\dfrac{\sqrt{x^3}}{\sqrt[3]{x}}$

Part F — Find the Error

19. Asked to evaluate $27^{2/3}$, a student wrote

$$27^{2/3} = \sqrt{27^3} = \sqrt{19\,683} \approx 140.3.$$

(a) Identify, in one sentence, the two things the student swapped.

(b) Give the correct value, and show the route that keeps the numbers smallest.

(c) The student's expression is not nonsense — write $\sqrt{27^3}$ as a power with a rational exponent.

20. A second student wrote $(-16)^{1/2} = -4$, reasoning that "the square root of 16 is 4, and the base was negative, so the answer is negative."

(a) Give *two* separate reasons the answer -4 is wrong.

(b) Change one thing about $(-16)^{1/2}$ — the brackets only — so that -4 *is* the correct answer.

(c) Keeping the base -16, is there any rational exponent that gives a real value? Give one, or explain why not.

21. Three more lines, each containing exactly one error. Name the mistake and write the correct answer.

(a) $a^{1/2} \cdot a^{1/3} = a^{1/6}$ (b) $\left(9x^4\right)^{1/2} = 9x^2$ (c) $8^{-2/3} = -4$

Part G — Stretch

22. A rational exponent can be undone by raising both sides to its *reciprocal*. Solve for x, and check each answer by substituting it back. Part (b) has more solutions than you might expect — find them all.

(a) $x^{3/2} = 27$ (b) $x^{2/3} = 25$ (c) $x^{-1/2} = \dfrac{1}{4}$ (d) $x^{5/3} = 32$

23. Nested radicals become easy once every root is an exponent. Convert, simplify, and give each answer as a single power of x. Assume $x > 0$.

(a) $\sqrt[3]{x\sqrt{x}}$ (b) $\sqrt{x\sqrt{x\sqrt{x}}}$ (c) $\dfrac{\sqrt[3]{x^2}\cdot\sqrt[4]{x}}{\sqrt{x}}$

24. Two exam-style items.

(a) *(Multiple Choice)* $\left(\dfrac{27}{8}\right)^{-2/3}$ is equal to

A. $\dfrac{9}{4}$ B. $\dfrac{4}{9}$ C. $-\dfrac{9}{4}$ D. has no meaning

(b) *(Numerical Response)* The value of $16^{3/4} + 64^{2/3} - 81^{1/2}$ is _____.
(Record your answer as a three-digit number.)

Lesson 6 — Extra Practice

Exponents Unit Review

Every variable represents a positive real number unless a question says otherwise. Write every final answer with positive exponents unless the question asks you to evaluate. No calculator anywhere in this set.

Part A — Simplifying with Every Law

1. Each expression below needs several exponent laws, together with a negative exponent *and* a rational exponent, before it is finished. Work the brackets first.

(a) $\dfrac{\left(8x^{3}y^{-6}\right)^{2/3}}{2x^{-1}y^{2}}$ (b) $\left(\dfrac{a^{-1/2}b^{3}}{a^{3/2}b^{-1}}\right)^{-2}$ (c) $\dfrac{16m^{5/2}n^{-3}}{\left(4m^{1/2}n\right)^{2}}$ (d) $\left(27p^{6}q^{-3}\right)^{-2/3}\cdot 9p^{2}q^{-1}$

2. Three more of the same kind. One of these answers keeps both a fractional exponent and a fraction in front — that is not a sign that you have gone wrong.

(a) $\left(\dfrac{x^{1/3}y^{-2}}{x^{-2/3}y}\right)^{3}$ (b) $\dfrac{\left(2a^{-2}b^{1/2}\right)^{-3}}{16a^{4}b^{-3}}$ (c) $\left(\dfrac{25c^{4}}{d^{-6}}\right)^{1/2}\div\left(5c^{2}d^{3}\right)^{-1}$

3. Each of the three expressions below collapses all the way to a *number*, so you can check your work by substituting. Simplify each one, then verify (a) by substituting $x=2$ and again $x=-3$.

(a) $\dfrac{\left(3x^{2}\right)^{3}\cdot x^{-4}}{27x^{2}}$ (b) $\left(\dfrac{4a^{3}b^{-1}}{a^{5}b^{-3}}\right)^{1/2}\cdot\dfrac{a}{2b}$ (c) $\dfrac{\left(16m^{8}\right)^{3/4}}{8m^{9}}\cdot m^{3}$

(d) **Explain** why two successful substitutions are good evidence but not a proof, and say what *does* prove the expression equals that number for every allowed value.

Part B — Evaluating Exactly

4. A fractional base, a fractional exponent and a negative exponent, all at once. Flip first, then take the root, then apply the power — the numbers stay small that way. Give every answer as an exact fraction in lowest terms.

(a) $\left(\frac{16}{81}\right)^{-3/4}$ (b) $\left(\frac{8}{125}\right)^{-2/3}$ (c) $\left(\frac{9}{4}\right)^{-5/2}$ (d) $\left(\frac{1}{32}\right)^{-3/5}$

5. Now with a zero exponent, a sum and a negative base mixed in. Evaluate exactly.

(a) $16^{-3/4} + 27^{2/3}$ (b) $\left(\frac{4}{9}\right)^{-1/2} - 8^0 \cdot 4^{-1}$ (c) $\frac{25^{3/2}}{5^{-1}}$ (d) $\left(32^{-2/5}\right)^{-1} + (-8)^{2/3}$

6. Exactly two of the following six powers have **no meaning** (as a real number). Identify them and say why; evaluate the other four exactly.

(a) $(-64)^{2/3}$ (b) $(-64)^{-1/2}$ (c) $-64^{1/2}$

(d) $(-32)^{-3/5}$ (e) $\left(-\frac{1}{16}\right)^{3/4}$ (f) $-\left(\frac{1}{16}\right)^{-3/4}$

Part C — Radicals and Rational Exponents

7. Rewrite each radical as a power first, then use the laws. Write each answer as a power with a positive exponent, and then as an entire radical.

(a) $\sqrt[3]{x^2} \cdot \sqrt[4]{x^3}$ (b) $\frac{\sqrt{y^5}}{\sqrt[3]{y^2}}$ (c) $\left(\sqrt[5]{z^2}\right)^{-3} \cdot z^2$ (d) $\left(\sqrt[3]{w^2}\right)^2 \div \sqrt[4]{w^5}$

8. Same job, but a coefficient now rides along. A coefficient outside the power stays outside the radical.

(a) $3\sqrt[3]{m^4} \times 2\sqrt{m}$ (b) $\dfrac{10\sqrt[4]{n^5}}{4n^{-1/4}}$ (c) $\left(8t^9\right)^{2/3} \div \left(4t^3\right)^{1/2}$

9. Working backwards. Find the missing exponent or index in each statement.

(a) $\left(\sqrt[4]{x^k}\right)^3 = x^9$ (b) $\dfrac{x^{7/3}}{x^k} = \sqrt[3]{x}$ (c) $\left(y^{2/5}\right)^k = \dfrac{1}{y^6}$ (d) $\sqrt[n]{b^6} = b^{3/2}$

10. Write both sides as powers of one common base, then equate the exponents. Not every answer is a whole number.

(a) $\dfrac{8^n}{2^5} = 2^7$ (b) $9^x = 27$ (c) $8^x = \dfrac{1}{4}$

(d) $\left(\dfrac{4}{9}\right)^x = \dfrac{27}{8}$ (e) $25^x \cdot 5^3 = 5^{-1}$ (f) $16^x = 8^{x+1}$

Part D — Scientific Notation in Context

11. For each row: write the given quantity in scientific notation, carry out the indicated operation using an exponent law, and report the result *both* in scientific notation and in standard decimal form.

(a) A square has side 0.0006 m. Find its area.

(b) A cube has edge 0.005 m. Find its volume.

(c) Square the number 80 000.

(d) Find the reciprocal of 0.0004.

12. A single bacterium has a mass of about 9.5×10^{-13} g.

(a) A colony holds 4.0×10^{8} of them. Find the colony's mass in scientific notation.

(b) How many bacteria have a combined mass of 1 g? Answer in scientific notation to the nearest hundredth.

(c) **Explain** which exponent law you used in (a) and which in (b), and why they are different.

13. Powers and roots of numbers already in scientific notation. For the roots, first rewrite the coefficient so the power of ten has an exponent the index divides evenly — for example $4.9 \times 10^{-7} = 49 \times 10^{-8}$. Give every answer in proper scientific notation.

(a) $(1.5 \times 10^{-4})^{2}$

(b) $\dfrac{(6.0 \times 10^{7})^{2}}{(2.0 \times 10^{-3})^{3}}$

(c) $(4.9 \times 10^{-7})^{1/2}$

(d) $(8.0 \times 10^{-9})^{1/3}$

(e) $(1.6 \times 10^{-7})^{1/2}$

(f) $(2.7 \times 10^{10})^{1/3}$

14. A cube-shaped tank holds 2.7×10^{-2} m^3 of liquid.

(a) Write the edge length as a power of the volume, then evaluate it exactly. Give the edge in both scientific notation and standard form.

(b) Find the total surface area of the tank, in scientific notation.

(c) A second tank is a cube of 8 times the volume. By what factor does the edge grow, and what is the new edge length?

Part E — Reasoning and Error Analysis

15. Mark each statement **true** or **false**. For every false one, give a specific counterexample and write the correct version of the statement.

(a) A negative exponent makes the value of a power negative.

(b) $\sqrt[n]{a^n} = a$ for every real number a and every natural number n.

(c) $a^{1/2} \cdot a^{1/2} = a$ for every real number a.

(d) If $a^m = a^n$, then $m = n$.

(e) $(-8)^{2/3} = 4$.

(f) $(5 \times 10^3)^2 = 25 \times 10^6$, written in scientific notation.

16. A student simplified $\dfrac{\left(2x^{-2}y^3\right)^{-2}}{8x^{-3}y^4}$ like this. Exactly one line is the first to go wrong.

Line 1: $\dfrac{-4x^4y^{-6}}{8x^{-3}y^4}$ Line 2: $-\dfrac{1}{2}x^{4-(-3)}y^{-6-4}$ Line 3: $-\dfrac{1}{2}x^7y^{-10}$ Line 4: $-\dfrac{x^7}{2y^{10}}$

(a) Name the first line that is wrong and say exactly what the student did.

(b) Rewrite that line correctly.

(c) Finish the simplification and state the correct final answer.

17. Where the rational-exponent rule comes from.

(a) Assume only that the product law holds for fractional exponents. Multiply $a^{1/3}$ by itself three times and use the result to **explain** why $a^{1/3}$ has to mean $\sqrt[3]{a}$.

(b) Use the quotient law in the same way to **explain** why a^0 has to equal 1, and why the law must be stated with $a \neq 0$.

(c) **Explain** why $\left(\sqrt[n]{a}\right)^m$ and $\sqrt[n]{a^m}$ give the same value, then evaluate $32^{4/5}$ by both routes and say which was easier.

18. Scientific notation is just the exponent laws in disguise.

(a) **Explain** why multiplying $(a \times 10^m)(b \times 10^n)$ may force you to renormalize the coefficient, even though the exponent step $m + n$ is never in doubt.

(b) If $a \times 10^m$ is written properly, what values can the coefficient of $(a \times 10^m)^2$ take before renormalizing, and for which a is renormalizing needed?

(c) Show that $(a \times 10^n)^{-1} = \dfrac{1}{a} \times 10^{-n}$, then explain why this is in proper scientific notation only when $a = 1$, and renormalize it for $1 < a < 10$.

Part F — Challenge

19. Order 2^{40}, 3^{25} and 5^{18} from least to greatest without a calculator. The three exponents share no common factor, so match them *two at a time*.

(a) 40 and 25 are both multiples of 5. Write 2^{40} and 3^{25} as fifth powers, and compare the two bases.

(b) 40 and 18 are both multiples of 2. Write 2^{40} and 5^{18} as squares, and compare those bases.

(c) State the full order, and **explain** why two pairwise comparisons were enough to settle all three.

20. Solve each equation by writing both sides as powers of a single base.

(a) $25^{\,x+1} = 125^{\,x}$ (b) $\left(\sqrt[3]{2}\right)^{x} = 16$ (c) $4^{x} \cdot 8^{\,x-1} = 2^{10}$ (d) $\left(\dfrac{1}{27}\right)^{x} = \sqrt{3}$

21. Consider $\left(\dfrac{a^{-2/3}b^{1/2}}{a^{1/3}b^{-1}}\right)^{-6}$, with $a > 0$ and $b > 0$.

(a) Simplify it, writing the answer with positive exponents.

(b) Evaluate the simplified form at $a = 8$ and $b = 4$.

(c) Find another pair $a, b > 0$ that gives the same value as (b). State the condition on a and b that makes it happen.

22. *(Multiple Choice)* The expression $\dfrac{\left(8x^6y^{-3}\right)^{2/3}}{2x^{-2}y^{-4}}$ simplifies to

A. $2x^6y^2$ B. $2x^2y^2$ C. $\dfrac{2x^6}{y^2}$ D. $4x^6y^2$

23. *(Numerical Response)* If $\dfrac{(6.0\times10^{-4})^2}{4.0\times10^{-11}} = 9.0\times10^n$, then the value of n is _____.
(Record your answer as a three-digit number.)